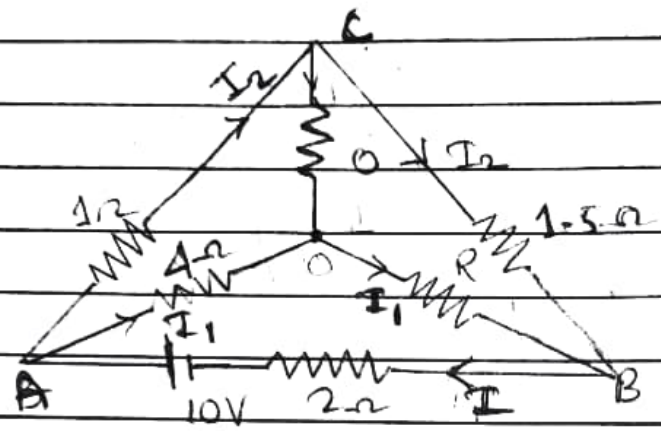
Q2. In the network shown calculate unknown value of R and the current flowing through it when the current in branch ⊙ is zero.



$I = I_1 + I_2$

In loop ABCA

3 / 13

$$10 - I(2) - 1(I_2) - (1.5)I_2 = 0$$

$$\cancel{3I_2 + 1.5}$$

$$2I + 2.5I_2 = 10$$

$$2I_1 + 4.5I_2 = 10 \quad \text{— (I)}$$

In loop AOBA

$$10 - 2I - 4I_1 - RI_1 = 0$$

In loop AOCA

$$-4I_1 + I_2 = 0$$

$$I_2 = 4I_1 \quad \text{— (II)}$$

Putting (II) in (I)

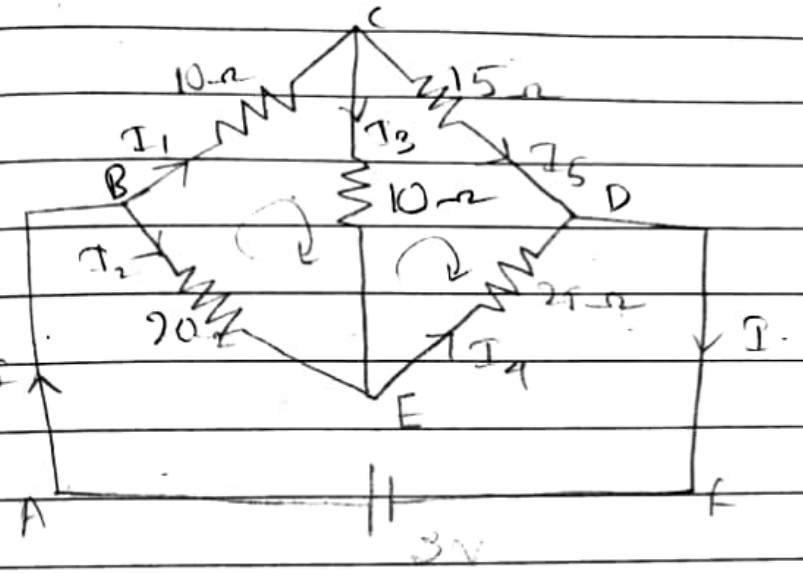
$$2I_1 + (4.5)(4I_1) = 10$$

$$20.5\,I_1 = 10$$

$$\boxed{I_1 = 0.5\ A}$$

$$\boxed{I_2 = 1\ A}$$

Q2) A cell of 2V with negligible internal resistance is connected across AC terminal. Calculate current in all branches.



In loop ABEDFA

$$20 I_2 + 25 I_4 = 2$$

$$2 - 20 I_2 + 25(I_2 + I_3) = 2$$

$$45 I_2 + 25 I_3 = 2 \quad \text{— (1)}$$

In loop I BCEB

$$-10 I_1 - 10 I_3 + 20 I_2 = 0$$

$$10(I_1 + I_3) = 20 I_2$$

$$(I_1 + I_3) = 2 I_2 \quad \text{— (2)}$$

$$I_1 - 2 I_2 + I_3 = 0$$

In loop CEDC

$$10 I_3 + 25 I_4 = 15 I_5$$

$$10 I_3 + 25(I_2 + I_3) = 5(I_1 - I_2)$$

$$10 I_3 + 25 I_2 + 25 I_3 - 5 I_1 + 5 I_3 = 0$$

$$150 I_3 + 25 I_2 - 5 I_1 = 0$$

$$10 I_3 + 5 I_2 - I_1 = 0 \quad \text{— (3)}$$

From above eqn

$$\begin{bmatrix} 0 & 45 & 25 \\ 1 & -2 & 1 \\ -1 & 5 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \frac{1}{-420} \begin{bmatrix} -25 & 11 & 3 \\ -325 & -25 & -45 \\ 95 & 25 & -45 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \frac{1}{-420} \begin{bmatrix} -50 \\ -650 \\ 190 \end{bmatrix}$$

$I_1 = 0.119$ A

$I_2 = 1.55$ A

$I_3 = -0.447$ A

Q4] A 50 Hz voltage of 215 V is impressed on a 50 Ω resistance.

i) Write the time equations for voltage and current

$V_m = 215$ V $R = 50\,\Omega$ $I_m = \frac{V_m}{R} = 4.3$ A

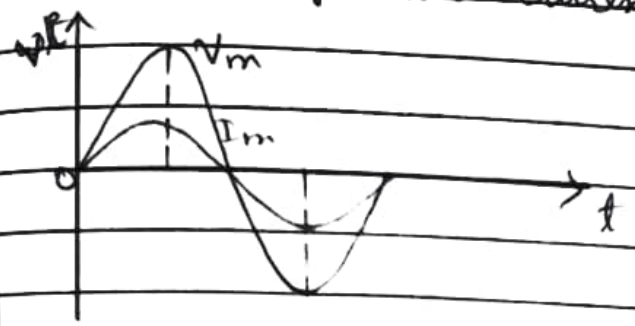
$V = V_m \sin \omega t$ $I = I_m \sin \omega t$

$f = 50$ Hz $\omega = 2\pi f = 2\pi(50) = 314.16$ rad/sec

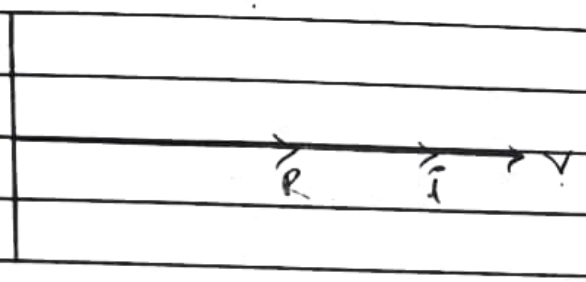
$V = 215\sqrt{2} \sin(314.16t) = 304.06 \sin(314.16t)$

$I = 4.3\sqrt{2} \sin(314.16t) = 6.08 \sin(314.16t)$

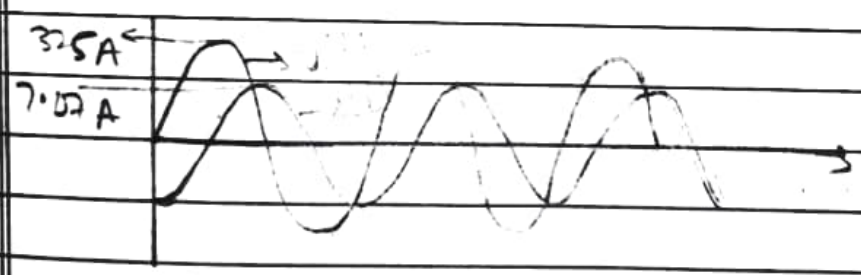
ii) Show voltage and current on time diagram



iii) Show voltage and current on phasor



5) A 60 Hz voltage source is impressed on an element and resultant wave form is shown in fig



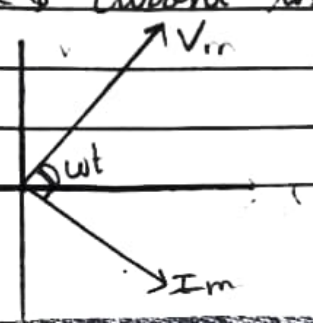
i) Write the time eq for voltage and current

$V_m = 325\ V \quad I_m = 7.07A \quad f = 60 \quad \omega = 377\ rad/s$

$V = V_m \sin(\omega t) = 325 \sin(377t)$

$I = I_m \sin(\omega t - \pi/2) = 7.07 \sin(377t - \pi/2)$

ii) Show voltage & current in phasor diagram



$Z_1 = 10 + 5j\ \Omega$ —(I)

$Z_2 = 8 + 6j\ \Omega$ —(II)

$$Z_{eff} = Z = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{(10+5j)(8+6j)}{10+8+5j+6j}$$

$$= \frac{80 + 40j + 60j + 30j^2}{18 + 11j}$$

$$= \frac{80 - 30 + 100j}{18 + 11j}$$

$$= \frac{50 + 100j}{18 + 11j}$$

$$Z = \frac{(50+100j)}{(18+11j)} \frac{(18-11j)}{(18-11j)}$$

$$= \frac{900 + 1800j - 550j - 1100j^2}{324 + 121}$$

$$= \frac{2000 + 1250j}{445}$$

$$Z = 4.5 + 2.8j$$

$$\phi = \tan^{-1}\left(\frac{2.8}{4.5}\right) = 31.9 \approx 32^\circ$$

$$\sqrt{4.5^2 + 2.8^2} = 5.3$$

$\therefore Z = 5.3\angle 32\ \Omega$ is total impendence

$$I_{total} = \frac{V_{total}}{Z_{total}} = \frac{200\angle 0^\circ}{5.3\angle 32^\circ} = [illegible]\angle 32^\circ$$

$Z_1 = 11.2\angle 26.6^\circ$

$Z_2 = \cancel{19\angle 18.4^\circ}\ 10\angle -36.86$

Since in parallel voltage remains same

$$I_1 = \frac{V}{Z_1} = \frac{200\angle 0^\circ}{11.2\angle 26.6} = 17.86\angle -26.6$$

$$I_2 = \frac{V}{Z_2} = \frac{200\angle 0}{10\angle 36.86^\circ} = 20\angle -36.86$$

$$\cos\phi_1 = \frac{R_1}{Z_1} = \frac{10}{11.2} = 0.89 \qquad \phi_1 = 26.56^\circ$$

$$\cos\phi_2 = \frac{R_2}{Z_2} = \frac{8}{10} \qquad \phi_2 = 36.87^\circ$$

$$\cos\phi = \left|\frac{4.5}{5.3}\right| = 0.849$$

$$\boxed{\phi = 31.9^\circ}$$

$$P_1 = I_1 V = 200 \times 17.86\angle -26.6$$
$$= 3572\angle -26.6$$

$$P_2 = I_2 V = 200 \times 20\angle -36.86 = 4000\angle -36.86$$

i) $I_1 = 17.86\angle -26.6^\circ$ A

$I_2 = 20\angle -36.86^\circ$ A

$I_{tot} = 37.7\angle -32^\circ$ A

ii) $\cos\phi_1 = 0.89$ $\qquad \phi_1 = 26.56^\circ$

$\cos\phi_2 = 0.8$ $\qquad \phi_2 = 36.87^\circ$

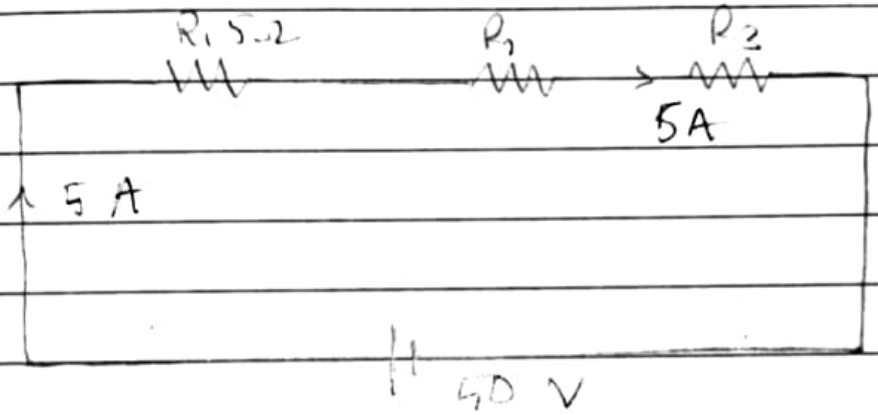
$\cos\phi = 0.85$ $\qquad \phi = 31.9^\circ$

iii) $P_1 = 3572\angle -26.6^\circ$ W

$P_2 = 4000\angle -36.6^\circ$ W

iv) $P = 7540\angle -32^\circ$ W

Q1) A resistance of $R_1 = 5\Omega$ is connected in series with an unknown resistance of R_2. This is connected in series with an unknown resistance of $R_3\ \Omega$ and circuit is the connected to 50 V dc supply. Calculate R_2 and R_3 is powers dissipated by R_3 is 150 W with 5 A passing through it



$V = 50\ V$

$I = 5\ A$

$$R_{eff} = \frac{V}{I} = \frac{50}{5} = 10\ \Omega$$

$$P_{R_3} = I^2 R_3 = 5^2 R_3$$

$$150 = 25 R_3$$

$$R_3 = \frac{150}{25} = 6$$

$$R_{eff} = R_1 + R_2 + R_3$$

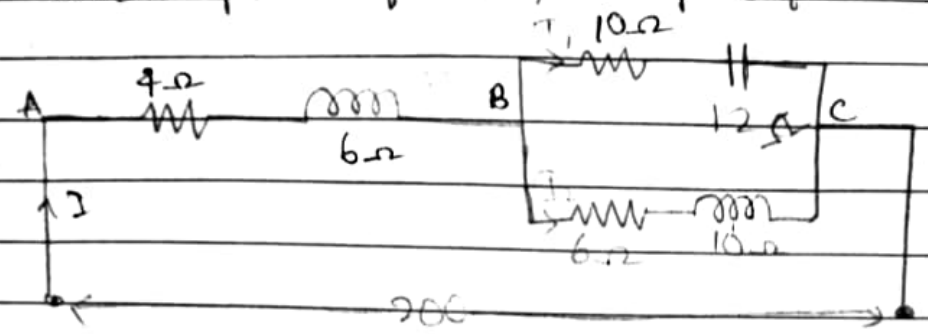
$$R_2 = R_{eff} - R_1 - R_2 = 10 - 5 - 6 = -1\ \Omega$$

$\therefore$ $R_1 = 5\ \Omega$

$R_2 = -1\ \Omega$

$R_3 = 6\ \Omega$

8) Determine the current drawn by the following circuit when a voltage of 200 V is applied across the . Draw the phasor diagram and find $I, I_1, I_2, V_{AB}, V_{BC}, P_1, P_2, P_{AB}$ overall power factor, total power



$V = 200$

$X_C = \frac{1}{2\pi f C}$

$X_L = 2\pi f L$

$V_L = I \times X_L$ — ①

$V_R = I \times R$ — ①

Voltage drop across $V_R = 4I$

$V_L = 6I$

$Z_{AB} = 4 + 6j = 7.21 \angle 56.31°$

$Z_{I_1} = 10 + 12j = 15.62 \angle +50.19°$

$Z_{I_2} = 6 + 10j = 11.66 \angle 59°$

$$Z_{BC} = \frac{Z_{I_1} \times Z_{I_2}}{Z_{I_1} + Z_{I_2}} = \frac{(10+12j)(6+10j)}{(10+12j)+(6+10j)}$$

$$= \frac{60 + 72j + 100j + 120j^2}{16 + 22j}$$

$$= \frac{180 + 28j}{16 + 22j} \cdot \frac{(16-22j)}{(16-22j)}$$

$$= \frac{-960 + 2756j + 1320j + 3784}{740}$$

$$= \frac{2824 + 4106j}{740}$$

$Z_{BC} = 3.8 + 5.54j = 6.72 \angle 55.5^\circ$

$Z_{tot} = Z_{AB} + Z_{BC} = 7.8 + 11.54j$
$= 13.9 \angle 55.9^\circ$

$I_{AB} = I_{BC} = I_{AC} = \dfrac{V}{Z_{tot}} = \dfrac{200 \angle 0^\circ}{13.9 \angle 55.9^\circ}$
$I = 14.4 \angle 55.9^\circ$

$V_{AB} = I_{AB} \times Z_{AB} = 14.4 \angle -55.9 \times 7.21 \angle 56.31^\circ$
$= 103.82 \angle 0.41^\circ$

$V_{BC} = I_{AB} \times Z_{BC} = 14.4 \angle -55.9 \times 6.72 \angle 55.5$
$= 96.77 \angle -0.4^\circ$

$I_1 = \dfrac{V_{BC}}{Z_{I_1}} = \dfrac{96.77 \angle -0.4^\circ}{15.62 \angle 50.19^\circ} = 6.2 \angle -50.59^\circ$

$I_2 = \dfrac{V_{BC}}{Z_{I_2}} = \dfrac{96.77 \angle -0.4^\circ}{11.66 \angle 59^\circ} = 8.3 \angle -59.4^\circ$

$P_1 = I_1 V_{BC} = 6.2 \angle -50.59^\circ \times 96.77 \angle -0.4^\circ = 600 \angle -50.99^\circ$

$P_2 = I_2 V_{BC} = 8.3 \angle -59.4^\circ \times 96.77 \angle -0.4^\circ = 803 \angle -59.8^\circ$

$P_{AB} = I_{AB} V_{AB} = 14.4 \angle -55.9^\circ \times 103.82 \angle 0.41^\circ$
$= 1495 \angle -55.49^\circ$

Total power $P = I_{AC} V = 14.4 \angle -55.9^\circ \times 200 \angle 0^\circ$
$P = 2880 \angle -55.9^\circ$

Overall power factor $\cos\phi = \dfrac{7.8}{13.9} = 0.56$

$\phi = 56^\circ$

$\therefore I = 14.4 \angle -55.9^\circ$

$I_1 = 6.2 \angle -50.59^\circ$

$I_2 = 8.3 \angle -59.4^\circ$

$V_{AB} = 103.82 \angle 0.41^\circ$

$V_{BC} = 96.7 \angle -0.4$

$P_1 = 600 \angle -50.99^\circ \ W$

$P_2 = 803 \angle -59.8^\circ \ W$

$P_{AB} = 1495 \angle -55.49^\circ \ W$

$P_{tot} = 2880 \angle -55.9^\circ \ W$

$\cos\phi = 0.56$

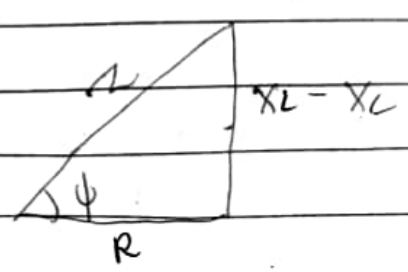
9) A voltage $v(t) = 100 \sin 314t$ is applied to series circuit consisting of $10\,\Omega$ resistance 0.0318 henry inductance and a capacitor of $63.6\,\mu F$. Calculate expression for its i) $i(t)$ ii) phase angle between voltage & current iii) power factor iv) active power consumed

$V_m = 100\,V \qquad R = 10\,\Omega \qquad L = 0.0318\,H$

$C = 63.6 \times 10^{-6}\,F \qquad \omega t = 314 \Rightarrow f = \frac{314}{2\pi} = 50\,Hz$

$X_L = 2\pi f L = 314 \times 0.0318 = 9.98 \simeq 10\,\Omega$

$X_C = \frac{1}{2\pi f C} = \frac{10^6}{314 \times 63.6} = 50.07 \simeq 50\,\Omega$



$$Z = \sqrt{(X_L - X_C)^2 + R^2} = \sqrt{-40^2 + 10^2} = 41.23\ \Omega$$

Now $I_m = \frac{V_m}{Z} = \frac{100}{41.23} = 2.425\ A$

$$\phi = \tan^{-1}\left(\frac{-40}{10}\right) = -75.96^\circ$$

i) $I(t) = I_m \sin(\omega t + \phi) = 2.42 \sin(314t - 75.96^\circ)$

ii) Phase angle between voltage & current $= \phi = 75.96^\circ$

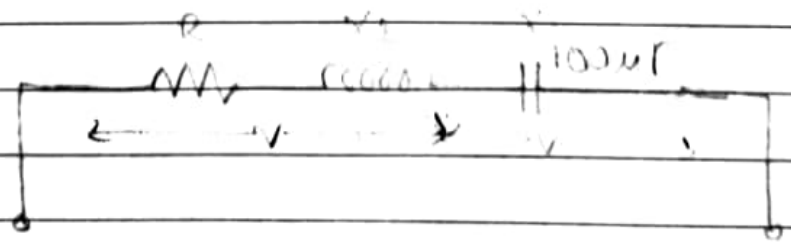
iii) $PF = \cos\phi = \left|\frac{R}{Z}\right| = \left|\frac{41.23}{10}\right| = 4.123$

iv) Active power consumed $= \frac{I_m V_m \cos\phi}{2}$

$= 100 \times 2.42 \times 4.123 / 2$

$P = 499\ W$

10) A coil of power factor 0.6 is in series with a 100 μF capacitor when connected to a 50 Hz supply. The Pd across the coil is same as Pd across the capacitor. Find resistance & Inductance.

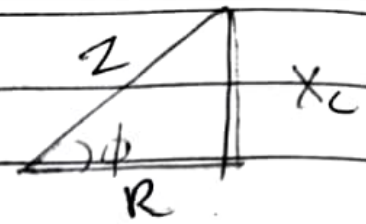


$\cos\phi = 0.6$

$C = 100\ \mu F \qquad f = 50\ Hz$

$$X_C = \frac{1}{2\pi f C} = \frac{10^6}{31400} = 31.85\ \Omega$$

$$\cos\phi = \frac{R}{Z} = 0.6$$



$$\phi = 53.13^\circ$$

$$\tan\phi = \frac{X_c}{R}$$

$$1.33 = \frac{X_c}{R} = \frac{31.85}{R}$$

$$R = \frac{31.85}{1.33} = 24\,\Omega$$

$$\cos\phi = \frac{R}{Z}$$

$$0.6 = \frac{24}{\sqrt{31.85^2 + X_c^2}}$$

$$0.36 = \frac{576}{1014.42 + X_c^2}$$

$$1014.42 + X_c^2 = 1600$$

$$X_c^2 = 585.58$$

$$X_c = 24.2$$

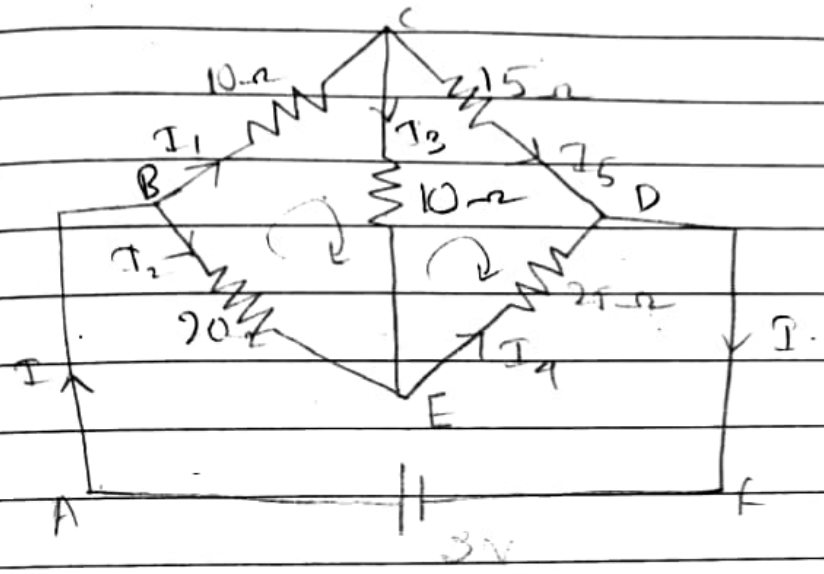
$$X_L = 2\pi f L = 314 \times 24.2$$

$$L = \frac{X_L}{2\pi f} = \frac{24.2}{314} = 0.077 \text{ Henry}$$

$\therefore$ $R = 24\,\Omega$

$L = 0.077$ henry

Q2) A cell of 2V with negligible internal resistance is connected across AC terminal. Calculate current in all branches.



In loop ABEDFA

$$20 I_2 + 25 I_4 = 2$$

$$2 - 20 I_2 + 25(I_2 + I_3) = 2$$

$$45 I_2 + 25 I_3 = 2 \quad \text{— (1)}$$

In loop I BCEB

$$-10 I_1 - 10 I_3 + 20 I_2 = 0$$

$$10(I_1 + I_3) = 20 I_2$$

$$(I_1 + I_3) = 2 I_2 \quad \text{— (2)}$$

$$I_1 - 2 I_2 + I_3 = 0$$

In loop CEDC

$$10 I_3 + 25 I_4 = 15 I_5$$

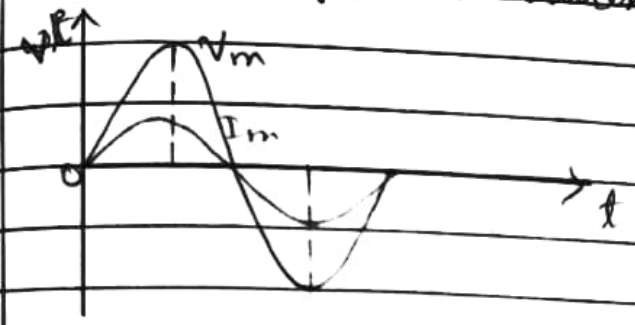
$$10 I_3 + 25(I_2 + I_3) = 5(I_1 - I_2)$$

$$10 I_3 + 25 I_2 + 25 I_3 - 5 I_1 + 5 I_3 = 0$$

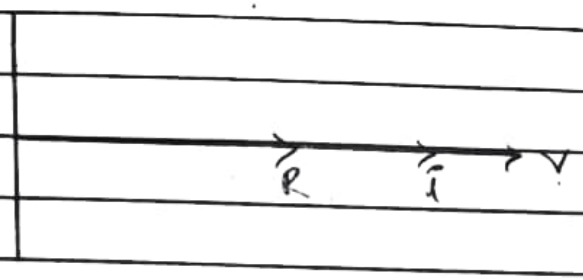
$$150 I_3 + 25 I_2 - 5 I_1 = 0$$

$$10 I_3 + 5 I_2 - I_1 = 0 \quad \text{— (3)}$$

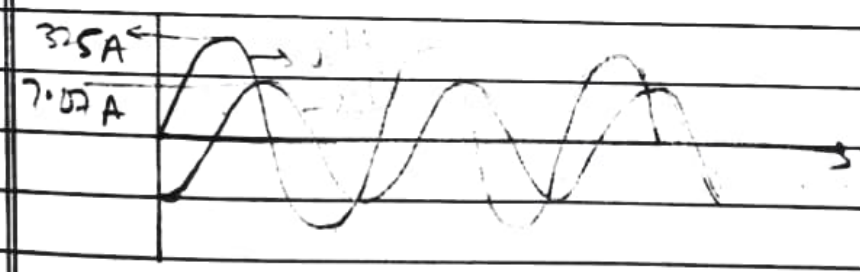
ii) show voltage and current on time diagram



iii) show voltage and current on phasor



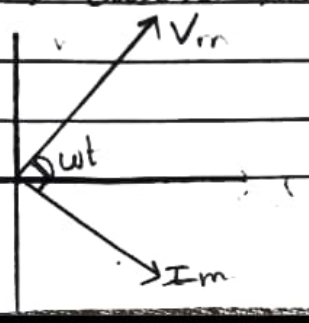
5) A 60 Hz voltage source is impressed on an element and resultant wave form is shown in fig



i) Write the time eq for voltage and current

$V_m = 325\ V \quad I_m = 7.07A \quad f = 60 \quad \omega = 377\ rad/s$

$V = V_m \sin(\omega t) = 325 \sin(377t)$

$I = I_m \sin(\omega t - \pi/2) = 7.07 \sin(377t - \pi/2)$

ii) show voltage & current in phasor diagram